Solution Requirements

Date	15 March 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium /Low)
1	Authentication	Users can securely register and log in using a username and password before accessing the system.	High
2	Authorization levels	Registered users can submit data and view their predictions. Administrators can manage datasets, users, and system settings.	High
3	External interfaces	The system interacts with the Flask web interface, CSV dataset, trained Linear Regression model, and visualization libraries (Matplotlib/Plotly).	Medium
4	Transactions processing	The system collects user health and lifestyle data, validates the input, processes it through the ML model, and stores the prediction results.	High

5	Reporting	The system generates the predicted well-being score, graphs, charts, and personalized recommendations through the dashboard.	High
6	Business rules, etc.	Only valid input values are accepted. Missing or incorrect values are rejected. Predictions are generated only after successful validation.	High
7	Compliance to laws or regulations	User information is stored securely and used only for educational purposes while maintaining privacy and confidentiality.	Medium
8	*Other*	The system allows users to compare previous predictions and monitor changes in well-being over time.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance & Speed	The system should process user input and generate predictions quickly.	Prediction generated within **2 seconds**
2	Scalability	The application should support multiple users and larger datasets without significant performance degradation.	Supports **500+ users** and large datasets

3	Security & Data Privacy	User data should be protected using secure authentication and encrypted storage.	Password hashing, HTTPS, secure session management
4	Reliability & Availability	The system should be available whenever users access it and recover from minor failures.	**99% uptime** during deployment
5	Usability & Accessibility	The application should have a simple, responsive, and user-friendly interface accessible from desktop and mobile devices.	Responsive UI with easy navigation
6	*Other*	The code should be modular, easy to maintain, and deployable on different operating systems.	Compatible with **Windows, Linux, and cloud deployment**